

Comparative Analysis Report

Comparative Analysis

Baseline

The baseline system measures how well a pre-trained instruction model can classify customer-support intent and follow strict JSON formatting without retrieved policy context. It is expected to be most vulnerable to formatting failures and unsupported policy claims.

Naive RAG

The RAG system retrieves relevant SOP passages and uses them as grounding context. This should reduce hallucinated refund windows, shipping timelines, password handling mistakes, and escalation errors. Retrieval quality depends on query wording and semantic similarity.

Hybrid RAG

The hybrid system uses a fine-tuned intent router to emit structured JSON before retrieval. The category and intent can be appended to the search query or used as a metadata filter, improving retrieval for messy, sarcastic, or ambiguous customer messages.

Recommendations

Use one fixed model ID, embedding model, dataset split, and inference configuration across all comparisons. Report format adherence, intent exact match, ROUGE/BLEU, consistency, and hallucination frequency on the held-out test split and adversarial subset only.